

Assignment #3

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2020
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Due Date: Wednesday February 19 by 5:00pm in my office

Number of points..... 100

Your score

Note: All problems are worth 100 points. Your final score will be the average value

Problem 3A: Review of the Ideal Gas and Maxwell Equations

Prove that:

$$\rho C_P \left(\frac{\partial T}{\partial P} \right)_H = \left(\frac{\partial \ln V}{\partial \ln T} \right)_P - 1$$

Using the formula you derived calculate the temperature change of an ideal gas that is undergoing an adiabatic expansion through a throttling valve.

First used is the final formula derived in class and the the thermal expansion coefficient into the equation.

$$dH = C_P dT + (1 - \beta T) V dP$$
$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$$

Then we plug the expansion coefficient into the equation.

$$dH = C_P dT + \left(1 - \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P T \right) V dP$$

Then we simplify the equation

$$dH = C_P dT + \left(1 - \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P T \right) V dP \rightarrow dH = C_P dT + \left(V - \left(\frac{\partial V}{\partial T} \right)_P T \right) dP$$

Then divide dP and keep H constant as shown in the given equation above $\rho C_P \left(\frac{\partial T}{\partial P} \right)_H = \left(\frac{\partial \ln V}{\partial \ln T} \right)_P - 1$.

$$\left(\frac{dH}{dP} \right)_H = C_P \left(\frac{dT}{dP} \right)_H + \left(V - T \left(\frac{\partial V}{\partial T} \right)_P \right)$$

Since H is constant, dH is equal to zero making the equation equal to zero.

$$0 = C_P \left(\frac{dT}{dP} \right)_H + V - T \left(\frac{\partial V}{\partial T} \right)_P$$

Then $C_P \left(\frac{dT}{dP} \right)_H$ is isolated and both sides are divided by V

$$C_P \left(\frac{dT}{dP} \right)_H \frac{1}{V} = (-V + T \left(\frac{\partial V}{\partial T} \right)_P) \frac{1}{V}$$

$$C_P \left(\frac{dT}{dP} \right)_H \frac{1}{V} = -1 + \frac{T}{V} \left(\frac{\partial V}{\partial T} \right)_P$$

Then the natural log is taken for the $\frac{T}{V} \left(\frac{\partial V}{\partial T} \right)_P$ and we take the partial derivative of it keeping pressure constant

To get
$$C_P \left(\frac{dT}{dP} \right)_H \frac{1}{V} = -1 + \left(\frac{\partial \ln(V)}{\partial \ln(T)} \right)_P$$

Then we assume the mass is one so using the formula for density

$\rho = \frac{m}{v} = \frac{1}{v}$ and we plug this into the equation

$$\boxed{\rho C_P \left(\frac{dT}{dP} \right)_H = \left(\frac{\partial \ln(V)}{\partial \ln(T)} \right)_P - 1}$$

For the second part of this problem we use

Z=1 for ideal conditions,

$$\rho C_P \left(\frac{\partial T}{\partial P} \right)_H = T \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P \partial P$$

$$\frac{\partial V}{\partial T} = \frac{R}{P}$$

$$\rho C_p \left(\frac{\partial T}{\partial P} \right)_H = \frac{RT}{VP} - 1$$

$$Z = \frac{PV}{RT} = 1$$

$$\frac{1}{Z} = \frac{RT}{VP}$$

Then plugging it into the equation

$$\rho C_p \left(\frac{\partial T}{\partial P} \right)_H = \frac{1}{Z} - 1$$

And since Z is equal to 1, that makes the equation $\frac{1}{1} - 1 = 0$

$$\rho C_p \left(\frac{\partial T}{\partial P} \right)_H = 0$$

And isolating the $\left(\frac{\partial T}{\partial P} \right)_H$

$$\boxed{\left(\frac{\partial T}{\partial P} \right)_H = 0}$$

Thus resulting in the temperature change of an ideal gas that is undergoing an adiabatic expansion through a throttling valve equaling to zero.

Prove that:

$$C_v \left(\frac{\partial P}{\partial \rho} \right)_S = C_p \left(\frac{\partial P}{\partial \rho} \right)_T$$

The adiabatic speed of sound is defined as $\vartheta^2 = \left(\frac{\partial P}{\partial \rho} \right)_S$ Using the formula you derived above

calculate the adiabatic speed of sound in the air at ambient temperature assuming that the air behaves as an ideal gas. Compare your result with values published in the open literature.

Using internal energy

$$U = U(T, V) \text{ and since } C_p = \left(\frac{\partial H}{\partial T} \right)_P \text{ and } C_v = \left(\frac{\partial U}{\partial T} \right)_V$$

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$

$$\text{(Equation 1) } dU = C_v dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$

$$\text{(equation 2) } dU = TdS - PdV \text{ then } TdS = dU + PdV$$

Then plugging in dU into equation 2

$$TdS = C_v dT + \left[T \left(\frac{\partial P}{\partial T} \right)_V - P \right] dV$$

And using the ideal gas law $PV=RT$, $\left(\frac{\partial P}{\partial T} \right)_V = \frac{R}{V} = \frac{P}{T}$ and so this is plugged into the equation above and dividing by T

$$\frac{TdS}{T} = C_v dT + \left(T \left(\frac{R}{V} \right) - P \right) dV$$

$$dS = \frac{C_v dT}{T} + \left(\frac{\partial P}{\partial T} \right)_V dV$$

So,

$$dS = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV$$

Then we can conclude,

$$\frac{C_v}{T} = \left(\frac{\partial S}{\partial T} \right)_V$$

A similar method is used for Enthalpy

$$H = H(T, P)$$

$$dH = \left(\frac{\partial H}{\partial T} \right)_P dT + \left(\frac{\partial H}{\partial P} \right)_T dP$$

$$dH = C_p dT + (1 - \beta T) V dP \quad (\text{final formula obtained from class})$$

Then plugging in the equation below for dH in the equation above,

$$T dS = dH - V dP$$

And what is obtained is,

$$T dS = C_p dT + (1 - \beta T) V dP - V dP$$

Then both sides are divided by T

$$\frac{T dS = C_p dT + (1 - \beta T) V dP}{T}$$

$$dS = \frac{C_p}{T} dT + \beta V dP$$

So,

$$dS = \left(\frac{\partial S}{\partial T} \right)_P dT + \left(\frac{\partial S}{\partial P} \right)_T dP$$

$$\frac{C_p}{T} = \left(\frac{\partial S}{\partial T} \right)_P$$

Using $S=S(P,T)$

$$dS = \left(\frac{\partial S}{\partial P} \right)_T dP + \left(\frac{\partial S}{\partial T} \right)_P dT$$

Let's assume delta S is equal to 0 meaning constant entropy. Also because it is constant in the final formula given in the beginning of the problem.

$$\left(\frac{\partial S}{\partial T} \right)_P dT = - \left(\frac{\partial S}{\partial P} \right)_T dP$$

Then divide both sides by dT keeping entropy constant still,

$$\text{Equation 3} \quad \left(\frac{\partial S}{\partial T} \right)_P = - \left(\frac{\partial S}{\partial P} \right)_T \left(\frac{dP}{dT} \right)_S$$

Using equation 3, let's assume $S=S(V,T)$ instead so changing variables we get

$$\text{Equation 4} \quad \left(\frac{\partial S}{\partial T} \right)_V = - \left(\frac{\partial S}{\partial V} \right)_T \left(\frac{dV}{dT} \right)_S$$

Using

$$\frac{C_p}{T} = \left(\frac{\partial S}{\partial T} \right)_P \quad \text{and} \quad \frac{C_v}{T} = \left(\frac{\partial S}{\partial T} \right)_V$$

We can divide to get $\frac{C_p}{C_v} = \frac{\left(\frac{\partial S}{\partial T} \right)_P}{\left(\frac{\partial S}{\partial T} \right)_V}$

So, using this and equations 3 and 4 we plug in,

$$\frac{C_p}{C_v} = \frac{- \left(\frac{\partial S}{\partial P} \right)_T \left(\frac{dP}{dT} \right)_S}{- \left(\frac{\partial S}{\partial V} \right)_T \left(\frac{dV}{dT} \right)_S}$$

Then what's left is

$$\text{Equation 5} \quad \frac{C_P}{C_V} = \left(\frac{\partial V}{\partial P} \right)_T \left(\frac{\partial P}{\partial V} \right)_S$$

Using the formula for density $\rho = m/v$. it can be changed to $m = \rho * v$

$$dm = \rho dV + V d\rho$$

Assuming constant mass,

$$0 = \rho dV + V d\rho$$

$$-\frac{dV}{V} = \frac{d\rho}{\rho}$$

This can also equal to ,

$$\text{Equation 6} \quad \partial V = \frac{-V}{\rho} \partial \rho$$

Then plugging in equation 6 to 5.

$$\begin{aligned} \frac{C_P}{C_V} &= \left(\frac{\partial \rho}{\partial P} \right)_T \left(-\frac{V}{\rho} \right) \left(\frac{\partial P}{\partial \rho} \right)_S \left(-\frac{\rho}{V} \right) \\ \frac{C_P}{C_V} &= \left(\frac{\partial \rho}{\partial P} \right)_T \left(\frac{\partial P}{\partial \rho} \right)_S \end{aligned}$$

Which then gives,

$$\boxed{C_V \left(\frac{\partial P}{\partial \rho} \right)_S = C_P \left(\frac{\partial P}{\partial \rho} \right)_T}$$

For part two of the problem, given $\vartheta^2 = \left(\frac{\partial P}{\partial \rho} \right)_S$ and

$\gamma = \frac{C_P}{C_V}$ representing the adiabatic constant

$$C_V \vartheta^2 = C_P \left(\frac{\partial P}{\partial \rho} \right)_T$$

$$\vartheta^2 = \frac{C_P}{C_V} \left(\frac{\partial P}{\partial \rho} \right)_T$$

$$\vartheta^2 = \gamma \left(\frac{\partial P}{\partial \rho} \right)_T$$

Using the ideal gas law $PV=nRT$ we can change the n to m as mass and isolate P

$$P = \frac{m}{V}RT = P = \rho RT \text{ since } \rho \text{ is } m/V$$

Then plugging that in for P

$$\vartheta^2 = \gamma \left(\frac{\partial \rho RT}{\partial \rho} \right)_T = \gamma RT$$

$$\vartheta = \sqrt{\gamma RT}$$

Ambient temperature of air is 25degrees C.

γ constant for air is $1.4 \frac{kJ}{kg^\circ K}$ found from

<http://hyperphysics.phy-astr.gsu.edu/hbase/thermo/adiab.html>

Molecular weight of air is 28.964g/mol found from

https://www.engineeringtoolbox.com/molecular-mass-air-d_679.html

$$\vartheta = \sqrt{\gamma RT} = \sqrt{1.4 \frac{kJ}{kg^\circ K} * 8.314 \frac{J}{mol^\circ K} * (298.15K) \left(\frac{1mol}{28.964g} \right) \left(\frac{1000g}{1kg} \right)}$$

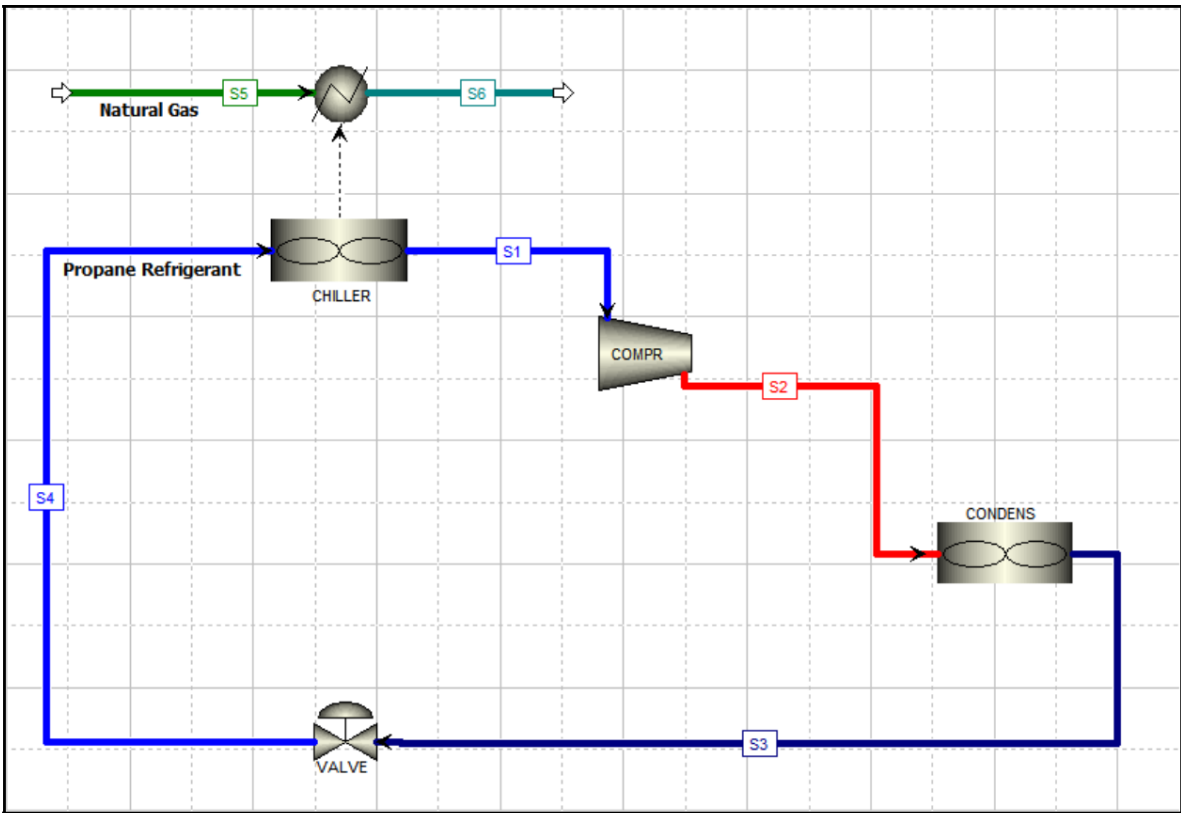
$$\vartheta = 346.14 \frac{m}{s}$$

Problem 3C: Natural Gas Cooling using Refrigerant Propane

We wish to cool {100 MMSCFD} of natural gas that consists primarily of methane from {25° C to – 37° C} using propane as the refrigerant. A process flowsheet diagram for this refrigeration system is shown below. Saturated liquid propane at {35° C} is expanded in a throttling valve down to a pressure of {1bar} and it is directed to a chiller, where the cooling of

the natural gas stream takes place. The refrigerant from the chiller is compressed and subsequently cooled in the condenser using cooling water. Using the **CHE301 Calculator** and assuming that propane and methane behave as ideal gases calculate the following:

1. The amount of refrigerant required in KMTA (assume a 90% operating factor)
2. The duties (in MW) of the chiller and the condenser.
3. The power requirement (in MW) of the compressor assuming
 - a. 75% isentropic efficiency
 - b. 75% polytropic efficiency
4. The COP (Coefficient of Performance) of the refrigeration cycle.
5. Compare your COP with the maximum possible value for this refrigeration cycle



Place all your answers in boxes for easy readability. Clearly state your assumptions and references for additional data you may need to answer the questions.

First, we convert 100MMSCFD

$$100\text{MMSCFD} * \frac{1180\text{m}^3/\text{hr}}{1\text{MMSCFD}} = 118,000 \frac{\text{m}^3}{\text{hr}}$$

Then the ideal gas law was used $PV=nRT$,

The standard conditions found in Ch.1 of Dr. C's notes is $T=60^\circ\text{F}$, $P=14.696\text{psia}$

$n=V_{tot} \frac{P}{RT}$ and converting the condition above from Fahrenheit to Rankine to fit with the R constant

with $P=14.69\text{psia}$, $R=10.7239 \frac{\text{psia ft}^3}{\text{lbmol R}}$, $T =$

519.67 R , V_{TOT} in $\frac{\text{ft}^3}{\text{hr}}$ to be able to get the n in $\frac{\text{lbmol}}{\text{hr}}$

So, first I converted the V_{TOT} to $\frac{\text{ft}^3}{\text{hr}}$

$$118,000 \frac{\text{m}^3}{\text{hr}} * 35.314 \frac{\text{ft}^3}{\text{m}^3} = 4167052 \frac{\text{ft}^3}{\text{hr}}$$

Then I plugged it into the equation to find n .

$$4167052 \frac{\text{ft}^3}{\text{hr}} * \left(\frac{14.69\text{psia}}{10.7239 \frac{\text{psia ft}^3}{\text{lbmol R}} * 519.67\text{ R}} \right) = 10,984.2476 \frac{\text{lbmol}}{\text{hr}}$$

Then I converted it to the desired units $\frac{\text{kmol}}{\text{s}}$

$$10,984.2476 \frac{\text{lbmol}}{\text{hr}} * \frac{453\text{mol}}{1\text{lbmol}} * \frac{1\text{kmol}}{10^3\text{mol}} * \frac{1\text{hr}}{60\text{min}} * \frac{1\text{min}}{60\text{s}} = \boxed{1.3835 \frac{\text{kmol}}{\text{s}}}$$

This flow rate is for the amount of methane gas going into stream 5 and we assume the heat exchanger doesn't change flowrate, so the flow rate for S6(output of heat exchanger) is 1.3835kmol/s .

Then I added components, comp1 is methane and comp2 is propane.

Stream 3 and 4(Throttling valve)

Then I assume flow is 0.3kmol/s starting in S1. This then results in S2,S3, and S4 equaling the flow rate of 0.3kmol/s because it's assumed a compressor, condenser, and throttling valve does not change the flowrate. Then the propane is assumed to be an ideal gas.

I focused on S3 the condenser.
The energy balance is found as

$$-\Delta H + Q + W = 0$$

We assume there's no work done and heat leaving.

$$\Delta H = 0$$

Then we can assume from this that $H_3 = H_4$ and focus on S3.

The temperature is given as 35 degrees Celsius as saturated liquid and pressure is found with $PSat(comp2, 35 \text{ degrees C})$. This gives 11.4361bar for stream4. $PSat$ function is used because the stream is saturated liquid.

Molar enthalpy is then found by using the function $H_3 = HIG(comp2, 35 \text{ degrees C}) - H_{vap}(comp2, 35 \text{ degrees C})$.

$$H_3 = -117027.6544 \text{ kJ/kmol}$$

Liquid enthalpy is the same number as H_3 because the flow of Stream 3 is saturated liquid.

$$H_{3L} = -117027.6544 \text{ kJ/kmol}$$

$$H_{3v} = \text{is found with } HIG(comp2, 35 \text{ degrees C}) = -103166.9836 \text{ kJ/kmol}$$

To find entropy again assuming ideal gas for propane,

For entropy

$$S_3 = SIG(comp2, 35 \text{ degrees C}, 11.4361 \text{ bar}) - H_{vap}(comp2, 35 \text{ degrees C}) / (35 \text{ degrees C} + 273.15) =$$
$$S_3 = -332.4169 \text{ kJ/kmol K}$$

(The 35 degrees C in Kelvin is divided in the entropy equation because the units for entropy has K in it while enthalpy doesn't)

Enthalpy is in kJ/kmol while Entropy is kJ/kmol K

$$S_{3v} = SIG(comp2, 35 \text{ degrees C}, 11.4361 \text{ bar}) = -287.4367 \text{ kJ/kmol K}$$

$$S_{3L} = -332.4169 \text{ kJ/kmol K since the stream is saturated liquid}$$

Since ΔH is equal to 0, H_4 needs to equal H_3 . So, for H_4 we make a guess for temperature.

The pressure of S4 is given as 1bar but I calculated it with PSat(comp2, guess temp) and the temperature is guessed as 10degrees Celsius. Then goal seek is used to get pressure equal to 1bar by changing the temperature and the temperature is found as -42.1525degreesC for stream4.

The formulas used from the previous column is dragged to the next column to give the H4L, H4v, and S4V, S4L.

A guess is used for the quality ,0.3, and the H4 is found by using The Lever Rule.

$$H4 = H4v(x) + (1 - x)H4L$$

Then I used secant method to make H3 and H4 difference equal to 0 by changing the quality to get 0.5390.

This gives H4= -117027.6544kJ/kmol

H4L= -127156.2608kJ/kmol

H4v= -108364.8958kJ/kmol

Then the entropy from the formulas used for Stream 3 gives S4= -367.8186kJ/kmolK=S4L

S4v= -286.4698kJ/kmolK.

STREAM1 and Stream 2(Compressor)

The temperature and pressure is assumed to be the same as S4 since the chiller doesn't change it.

The same formulas used for entropies and enthalpies are used to find the ones for this stream assuming that S4 propane is an ideal gas.

Also, the energy balance for the compressor is and we assume no heat is released or put on the system

$$-\Delta H + Q + W = 0$$

$$\Delta H = W$$

$$W = \dot{m}(H2 - H1)$$

Also, we assume constant entropy so $\Delta S = 0$

meaning $S1 = S2$.

T1=-42.1525degreesC

P1= 1bar

H1= HIG(comp2, -42.1525degreesC) =-108364.8957kj/kmol, same for H1v since the stream is assumed to be vapor since compressors take in vapor.

$$H1L==HIG(\text{comp2}, -42.1525\text{degreesC})- H_{\text{vap}}(\text{comp2}, -42.1525\text{degreesC})= -127156.2607\text{kJ/kmol}$$

$S1=SIG(\text{comp2}, -42.1525\text{degreesC}, 1\text{bar})= -286.4698\text{kJ/kmolK}$ which is the same for $S1V$ since it's all vapor

$$\text{For } S1L==SIG(\text{comp2}, -42.1525\text{degreesC}, 1\text{bar})-H_{\text{vap}}(\text{comp2}, C17)/(C17+273.15)= -367.8186\text{kJ/kmolK}$$

For Stream 2, since the condenser is assumed to have constant pressure, so $P3=11.4261\text{bar}=P2$.

Also, the compressor is assumed to have constant entropy so $\Delta S = 0$ meaning $S2 = S1$

Since $S1=-286.4698\text{kJ/kmolK}$ then $S2$ has to equal it so I made a guess for temperature for Stream 2 and I dragged the formulas from $S1$ to $S2$ column and so I used goal seek to make entropy of $S2$ equal to entropy of $S1$ by changing the temperature of stream 2.

Then obtained is $T2= 38.9219\text{degreesC}$

Then to find Q we use the formula obtained from the energy balance done above

$$W = \dot{m}(H2 - H1)$$

$$W = \dot{m}(H2 - H1)$$

But since flow rate isn't the exact one yet just the formula is found until used later.

CHILLER, S5, S6

Then for the chiller, heat is going out of the system since the chiller cools fluids and releases heat to the surroundings. As found <https://northslopechillers.com/blog/how-chiller-systems-work/>

For the streams of 5 and 6, we assume 1bar . The temperatures are given and to find the heat going into the heat exchanger, it has to be the same amount as the chiller heat out which was calculated by taking

$$Q(\text{out}) = \dot{m}(H4 - H1)$$

$$\text{So I get } Q_5 = 0.3\text{kmol/s}(-117027.6544\text{kJ/kmol}- -108364.8957\text{kJ/kmol}) = -2598.82\text{kJ/s}$$

Then I calculated the enthalpy of $S5$ and $S6$ by using the temperatures given and doing

$$H5=HIG(\text{comp1}, 25\text{degreesC}) = -74901.9680\text{kJ/kmol}=H5$$

$$\text{And the same for stream 6 } H6 =HIG(\text{comp1}, -37\text{degreesC}) = -76998.5652\text{kJ/kmol}=H6$$

And I calculate the heat which the heat exchanger is receiving which is the same formula but different flow rate and enthalpies.

$$Q = \dot{m}(H_6 - H_5)$$

$$Q_6 = 1.3835 \text{ kmol/s} (-76998.5652 \text{ kJ/kmol} - -74901.9680 \text{ kJ/kmol}) = -2900.6422 \text{ kJ/s} = Q_6$$

Then I found the actual flow of the streams besides S5 and S6 by using extended secant and setting the difference between Q_6 and Q_5 to 0 by changing the flow of S4.

This gets the flow to

0.3348 kmol/s for stream4 which should be the same for S3, S2 and S1 since we assume the compressor, condenser, and throttling valve doesn't change the flow rate.

$$0.3348 \text{ kmol/s} * \frac{60 \text{ s}}{1 \text{ min}} * \frac{60 \text{ min}}{1 \text{ hr}} * \frac{10^3 \text{ mol}}{1 \text{ kmol}} * \frac{1 \text{ lbmol}}{453 \text{ mol}} = 2660.6622 \text{ lbmol/hr}$$

Then plug into $V_{\text{tot}} = nRT/P$ to get ft cubed/hr

$$V_{\text{tot}} = \frac{nRT}{P} = \frac{2660.6622 \text{ lbmol/hr} (10.7239 \frac{\text{psia ft}^3}{\text{lbmol R}}) (519.67 \text{ R})}{14.69 \text{ psia}} = 1009365.242 \frac{\text{ft}^3}{\text{hr}}$$

$$1009365.242 \frac{\text{ft}^3}{\text{hr}} * \frac{1 \text{ m}^3}{35.314 \text{ ft}^3} = 28724.11047 \frac{\text{m}^3}{\text{hr}}$$

$$28724.11047 \frac{\text{m}^3}{\text{hr}} * \frac{1 \text{ MMSCFD}}{1180 \text{ m}^3/\text{hr}} = 24.342 \text{ MMSCFD of flow for the S1, S2, S3, and S4}$$

Then to find the Work on the compressor is

$$W = \dot{m}(H_2 - H_1)$$

$$W_2 = 0.3348 \text{ kmol/s} (-102867.1396 - -108364.8957 \text{ kJ/kmol})$$

$$= 1840.6652 \text{ kJ/s (kW)} = W_2$$

For the condenser,

$$Q_3 = \dot{m}(H_3 - H_2) = 0.3348 \text{ kmol/s} (-117027.6544 \text{ kJ/kmol} - -102867.1396 \text{ kJ/kmol})$$

$$Q_3 = -4740.9828 \text{ kJ/s (kW)}$$

		vap	vap	liquid satliopropane	w/L		methane gas	methane gas
Conditions and Properties	Units	s1	s2(Compressor)	s3(condensor)	s4(throttling valve)	Chiller	s5	s6(heatexch)
Molar Flow	kmol/s	0.3348	0.3348	0.3348	0.3348		1.3835	1.3835
Mass Flow	kg/s							
Volume Flow	m3/s							
Temperature	C	-42.1525	38.9219	35.0000	-42.1525		25.0000	-37.0000
Pressure	bar	1.0000	11.4361	11.4361	1.0000		1.0000	1.0000
Vapor Fraction[Quality]		1.0000		0.0000	0.5390			
Molecular Weight	kg/kmol							
Molar Enthalpy	kJ/kmol	-108364.8957	-102867.1396	-117027.6544	-117027.6544			
Vapor Molar Enthalpy		-108364.8957	-102867.1396	-103166.9836	-108364.8958		-74901.9680	-76998.5652
Liquid Molar Enthalpy		-127156.2607	-116390.3429	-117027.6544	-127156.2608			
Molar Entropy	kJ/(kmol K)	-286.4698	-286.4698	-332.4169	-367.8186			
Vapor Molar Entropy		-286.4698	-286.4698	-287.4367	-286.4698			
Liquid Molar Entropy		-367.8186	-329.8034	-332.4169	-367.8186			
Molar Density	kmol/m3							
Vapor Molar Density								
Liquid Molar Density								
Enthalpy Flow	kJ/s	-36280.8918	-34440.2266	-39181.2093	-39181.5340			
Duty[Power]	kJ/s[kw]		1840.6652	-4740.9828		-2900.6422		-2900.6422

QUESTIONS

1. $0.3348 \frac{\text{kmol}}{\text{s}}$ is the flow and to change to *KMTA* is 1000 metric tons per year from <http://www.freepatentsonline.com/7727490.html>

1000 metric tons per year is = 0.0316887646 kg/s

$$0.3348 \frac{\text{kmol}}{\text{s}} * \frac{44.1 \text{ kg}}{\text{kmol}} = 14.7647 \frac{\text{kg}}{\text{s}} * \frac{1000 \text{ metric tons per year}}{0.0316887646 \text{ kg/s}}$$

$$= 465,927.9144 \text{ metric tons per year}$$

$$465,927.9144 \text{ metric tons per year} * 0.9 = 419335.123 \text{ metric tons per year}$$

$$0.3348 \frac{\text{kmol}}{\text{s}} * 0.9 = 0.301 \frac{\text{kmol}}{\text{s}}$$

2. Duties of chiller and condenser (with 90% operating factor included)

$$Q(\text{out}) \text{ condenser} = n_4(h_3 - h_2) = 0.301 \text{ kmol/s} (-117027.6544 - -102867.1396 \text{ kJ/kmol})$$

$$= -2607.49 \text{ kW} * (0.001 \text{ MW/1kW}) = -4.26 \text{ MW}$$

$$Q(\text{out}) \text{ chiller} = n_1(h_4 - h_1) = 0.301 \text{ kmol/s} (-117027.6544 - -108364.8957 \text{ kJ/kmol})$$

$$= -631.08 \text{ kW} * (0.001 \text{ MW/1kW}) = -2.607 \text{ MW}$$

3.

a) Isentropic Efficiency

$$0.75 = W_{\text{ideal}} / W_{\text{real}} = 1840.6652 / W_{\text{real}} = 0.75$$

$$W_{\text{real of compressor}} = 2454.22 \text{ kW} * (0.001 \text{ MW/1kW}) = 2.454 \text{ MW}$$

b.) Polytropic Efficiency

From the formula found on Dr.C's notes for CH.2

Polytropic Efficiency for compressor (75%)

$$kW = W_s = \frac{\gamma}{\gamma - 1} RT_1 \left[\left(\frac{P_2}{P_1} \right)^{\frac{\gamma - 1}{\gamma}} - 1 \right] \dot{m}$$

First to find $\gamma = C_p / C_v$ using https://www.engineeringtoolbox.com/specific-heat-capacity-propane-Cp-Cv-isobaric-isochoric-d_2060.html for propane at -42.1525 degrees C

$$C_p = 0.0634 \text{ kJ/kmolK}$$

$$C_v = 0.0536 \text{ kJ/kmolK}$$

$$\gamma = 1.18$$

From the streams 1 and 2, $T_1 = -42.1525 \text{ degreesC} = 230.998 \text{ K}$, $P_2 = 11.43 \text{ bar}$, $P_1 = 1 \text{ bar}$

$$W_s = \left(\frac{1.18}{1.18 - 1} \right) 8.314 \text{ kJ/kmolK} (230.998 \text{ K}) \left[\left(\frac{11.43}{1} \right)^{\frac{1.18 - 1}{1.18}} - 1 \right] 0.3348 \text{ kmol/s}$$

$$= 1897.22 \text{ kW}$$

For 75%

$$W_s = 1897.22 \text{ kW} / 0.75 = 2529.63 \text{ kW} * (0.001 \text{ MW/1kW}) = 2.53 \text{ MW}$$

But since the operating factor is 90% we need to change the flow to 0.301 kmol/s and use the formula above

$$\begin{aligned}
 W_s &= \left(\frac{1.18}{1.18 - 1} 8.314 \text{ kJ/kmolK} (230.998 \text{ K}) \left[\left(\frac{11.43}{1} \right)^{\frac{1.18-1}{1.18}} - 1 \right] \right) 0.301 \text{ kmol/s} \\
 &= 1705.32 \text{ kW} \\
 W_s &= 1705.32 \text{ kW} / 0.75 = 2273.76 \text{ kW} = 2.27 \text{ MW}
 \end{aligned}$$

4. COP of refrigeration cycle

For 75% isentropic compressor

$$\text{COP} = Q(\text{chiller in}) / W_{\text{net}} = 2.607 \text{ MW} / (2.454 \text{ MW}) = 1.06 = \text{COP}$$

For polytropic compressor

$$\text{COP} = Q(\text{chiller in}) / W_{\text{net}} = 2.607 \text{ MW} / (2.27 \text{ MW}) = 1.148 = \text{COP}$$

5. Max COP is the COP under ideal conditions

$$\text{COP} = Q(\text{chiller in}) / W_{\text{net}} = (2900.6422 \text{ kW}) / (1840.6652 \text{ kW}) = 1.576 = \text{COP}_{\text{max}}$$